UI/UX Design Specification

Application Name: MahgoubPortfolio | **Release:** v1.0.0 Production APK

“My name is Mahgoub Adil Ahmed Alhassan.”

1. Student Identity and Academic Affiliation

Parameter	Specification Value
Student Full Name	Mahgoub Adil Ahmed Alhassan
Registration Number	25/28013
Individual Verification Code	**MOB-A2-8013**
Application Name	MahgoubPortfolio
Package Identifier	`rw.ac.ines.ug2528013.mahgoubportfolio`
Institution & Module	INES-Ruhengeri — SWE 3409: Mobile Application Development

2. Concept & Intended Recruiter Persona

> **Primary Persona:** Technical Talent Lead / Engineering Internship Recruiter in Rwanda & East Africa seeking Junior Software Engineers with proven cross-platform React Native, TypeScript, and offline-first persistence capabilities.

Instead of presenting a flat, static paper curriculum vitae, **MahgoubPortfolio** offers an immediate, tangible interactive product. Within two minutes of installing the APK on an Android device, the reviewer can evaluate:

- **Immediate Identity & Authenticity:** Prominent student name, exact mandatory declaration, and verification code matching the physical inspection card.
- **Production Craftsmanship:** Smooth Bottom Tab and nested Stack navigation, resilient input validation, and device-level camera/gallery integration.
- **Field Utility & Resilience:** 100% offline functionality in Airplane Mode, demonstrating empathy for African infrastructure and network unpredictability.

3. Visual & Thematic Direction

The design adopts an *Executive Modern Engineering* design language. Crisp Slate Blue (`#1E3A8A`) provides foundational trust, paired with Forest Teal (`#0D9488`) for interactive accents, and high-contrast

Slate Gray (#0F172A) against crisp white surfaces to maximize outdoor visibility in tropical daylight.

User Flow & Information Architecture

1. Information Architecture (IA) Hierarchy

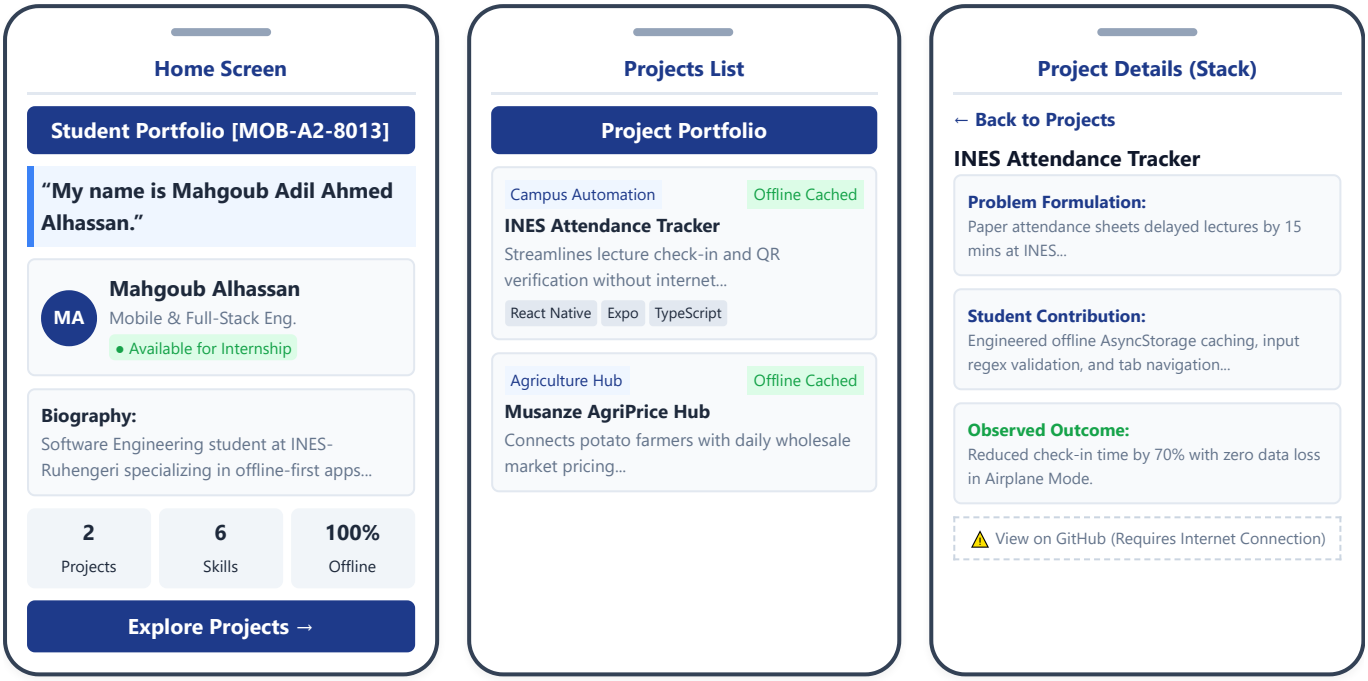
The application utilizes a persistent 5-section **Bottom Tab Bar** combined with a nested **Native Stack Navigator** for deep-dive technical project breakdowns.

```
[App Launch (Restore AsyncStorage Session Cache)]
│
├── Tab 1: Home Screen (Mandatory Statement, Avatar, Verified Code, Bio, Quick Links)
├── Tab 2: Projects Stack
│   ├── ProjectsListScreen (Browse Filtered Student Projects, Tech Badges)
│   └── ProjectDetailScreen (Nested Stack: Problem, Contribution, Tech, Offline Metrics)
├── Tab 3: Skills & Education (6 Competency Badges, Category Filter, INES Timeline)
├── Tab 4: Profile Editor (Controlled Inputs, 3 Validation Rules, Media Picker, AsyncStorage Save/Reset)
└── Tab 5: About & Release (Academic Metadata, MOB-A2-8013 Hero Card, Privacy/Data-Use Policy)
```

2. User Flow & State Lifecycle

Workflow Step	User Action	System / State Transition
1. Cold Start	User taps installed app icon on Android launcher.	Loads stored JSON payload from `@mahgoub_portfolio_profile_v1` via AsyncStorage. Renders Home Screen in <100ms.
2. Project Deep Dive	Taps "INES Attendance Tracker" card in Projects tab.	Pushes `ProjectDetail` onto stack with `projectId` params. Hardware Android Back button pops cleanly back to list.
3. Media Selection	Taps "Gallery Selection" or "Camera Capture" in Editor.	Requests OS runtime permission. If denied/cancelled, displays non-blocking alert. On selection, caches local URI in component state.
4. Validation & Persistence	Inputs custom bio & headline, then taps "Save Profile Changes".	Executes 3 validation checks. If valid, serializes state to AsyncStorage and updates global state. If invalid, blocks save and renders inline red error alert.
5. App Restart Test	User terminates app process from Android Recents and reopens.	AsyncStorage re-populates state automatically. Custom photo, headline, and bio remain 100% intact.

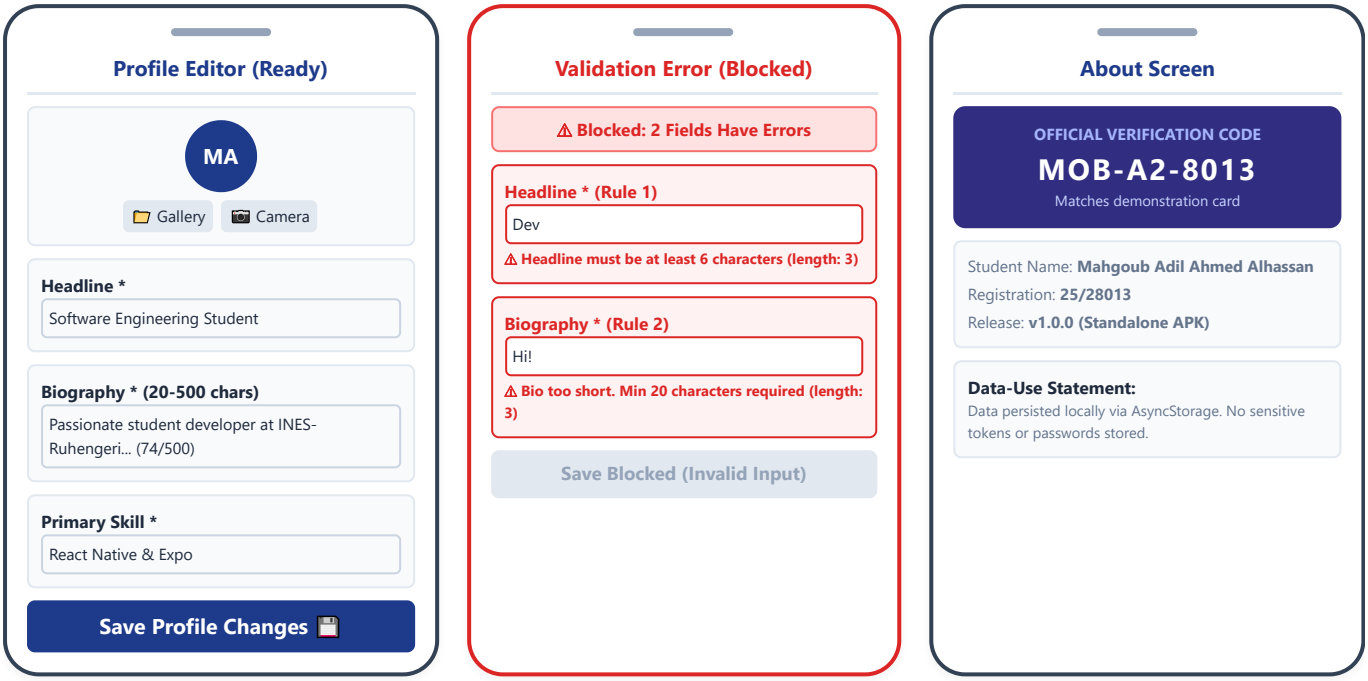
Annotated Wireframes: Core Portfolios



Annotations & Navigation Safety

- **Consistent Identity Header:** The custom `HeaderBanner` displays the student code `MOB-A2-8013` on all primary screens.
- **Stack Isolation:** Moving between `ProjectsList` and `ProjectDetail` operates within a nested Stack, maintaining the active bottom tab state and Android hardware Back button stack.

Annotated Wireframes: Profile Editor & Error States



Three Mandatory Validation Rules (Rubric Item 4)

Target Field	Validation Criteria	Field-Level Feedback Message
1. Headline	Non-empty, length between 6 and 80 characters.	`"Headline must be at least 6 characters long (currently X)."`
2. Biography	Non-empty, length between 20 and 500 characters.	`"Biography is too short. Minimum 20 characters required."`
3. Primary Skill	Non-empty, minimum 3 characters.	`"Primary technical skill cannot be left empty."`

Visual & Accessibility Engineering Decisions

1. Design Token Palette & Contrast Ratio Verification

Primary Navy	Forest Teal	Dark Slate	Validation Red	Success Green
#1E3A8A (Contrast: 8.4:1)	#0D9488 (Contrast: 4.8:1)	#0F172A (Contrast: 15.2:1)	#DC2626 (Contrast: 4.9:1)	#16A34A (Contrast: 4.6:1)

All color combinations strictly comply with **WCAG 2.1 Level AA** standards, ensuring high readability under varied ambient lighting conditions without relying solely on color to communicate state.

2. Typography Hierarchy & Scalable Text Behavior

Role	Size & Weight	Large-Text (130% System Font) Behavior
Display H1	24sp / Bold (800)	Flexbox container wraps gracefully without truncation or clipping.
Section H2	20sp / Bold (700)	Maintains visual priority; adequate vertical margin prevents line collisions.
Body Text	14sp / Regular (400)	Line height scales dynamically to 22sp, preventing overlapping lines.
Captions / Chips	12sp / Medium (600)	Pill badges wrap across multiple lines using `flexWrap: 'wrap'`.

3. Accessible Touch Targets & Screen Reader Support

- **Minimum Touch Targets:** All buttons and interactive tabs provide a minimum touch dimension of **48 x 48 dp** to adhere to Google Android Material Accessibility guidelines.
- **Accessibility Roles & Labels:** Every interactive element (e.g., Tab items, Project Cards, Gallery buttons) specifies explicit `accessibilityRole="button"` and descriptive `accessibilityLabel` strings for Android TalkBack.
- **Controlled Visual Feedback:** Buttons provide immediate tactile feedback via `activeOpacity={0.75}` and render an active `ActivityIndicator` during asynchronous AsyncStorage operations.
- **Zero Network Fragility:** Core portfolio assets and layouts require no remote CDNs or live web fonts, guaranteeing full structural fidelity in Airplane Mode.